

Sprint 3: Identification of potential ragas for aiding in meditation

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Introduction to Raga Meditation

Rooted in traditional sound healing and *Nada Yoga*, raga meditation utilizes modal melodic frameworks engineered to balance emotional states (*rasas*) and harmonize the nervous system. By emphasizing the slow, unmeasured development of the *alap*, raga meditation quiets intrusive thoughts, aligns respiration, and anchors cognitive awareness into contemplative stillness.

Technical Structure of Early-Morning and Morning Ragas

The musical architecture of the first and second *prahars* (4:00 AM – 10:00 AM) uniquely facilitates inward stabilization:

- **Raga Bhairav:** Employs flattened second (*Komal Rishabh*) and sixth (*Komal Dhaivat*) notes with slow, deliberate microtonal oscillations (*andolan*). These heavy pitch modulations eliminate abrupt melodic leaps, providing a neuro-acoustic anchor that dampens cortical hyperarousal into reverent stillness.
- **Raga Lalit:** Omits the stabilizing fifth (*Pancham*) while juxtaposing natural (*Shuddha*) and sharp (*Teevra*) fourths (*Madhyam*). This delicate chromatic proximity creates subtle introspective tension that gently resolves into the tonic (*Shadja*), clearing sleep inertia without causing agitation.
- **Raag Todi (Miyan ki Todi):** Features lowered *Rishabh*, *Gandhar*, and *Dhaivat* paired with sharp *Madhyam*. This dense cluster of flattened intervals exerts a weighty melodic gravity (*gambhira*) that tempers sympathetic arousal and sustains alert calm.
- **Raag Bilaskhani Todi:** Relies on unbroken descending glides (*meend*), eliminating percussive attacks to smooth auditory transitions, promote alpha-wave coherence, and facilitate emotional catharsis.

Rationale for considering Morning Ragas for the Project

These specific early-morning and morning ragas are being considered for the project because they target the critical transition between restorative sleep and daytime mental engagement. In the broader framework of how musical ragas affect human psychological well-being and daily life, morning hours mark a sensitive neuroendocrine shift accompanied by rising cortisol, changing circadian rhythms, and sleep inertia. Rather than introducing rhythmically aggressive or stimulating structures that can trigger acute sympathetic stress, these ragas utilize soothing flattened intervals (*komal swaras*), gradual microtonal oscillations, and fluid glissandos. This acoustic profile facilitates gentle cognitive awakening, suppresses anxiety, and establishes a balanced parasympathetic baseline, making them an ideal, standardized acoustic intervention for evaluating meditative efficacy.

References

- [1] Insight Timer, “Raga Meditation: Techniques and Sound Healing,” <https://insighttimer.com/techniques/sound-healing/raga-meditation>.
- [2] Deepika Tiwari, *et al.*, “Importance Of Music And Ragas In Human Life,” *Educational Administration: Theory and Practice*, vol. 30, no. 1, pp. 3670–3676, 2024, doi: [10.53555/kuey.v30i1.7413](https://doi.org/10.53555/kuey.v30i1.7413).